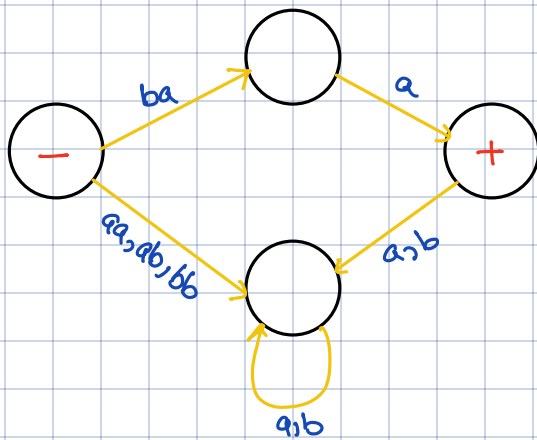
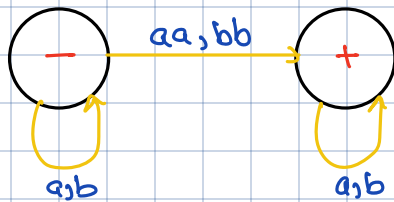


↳ In previous chapters, we built a FA that accepts the word baa but it required five states as it reads only one letter at a time. What if we could read more than one letter. For same language:



↳ Language that contains double letters



Here,

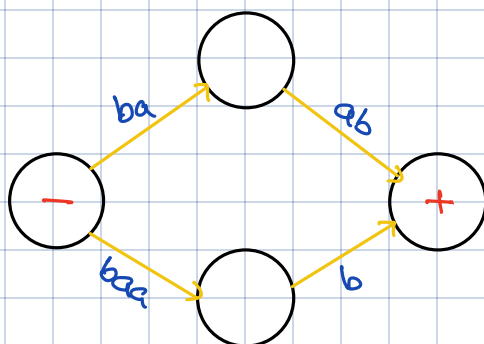
b a a is not acceptable despite having double letters. So, there are two different paths for baa. Also, b a a is acceptable but the machine crashes.

↳ When an input string that has not been completely read reaches a state that it cannot leave because there is no outgoing edge to follow.

↳ To allow machines that read more than one letter at a time, we have to change our definition of acceptance.

↳ An input is accepted if there is some way to reach the final state.
 ↳ We ignore failures

↳ Consider another example,



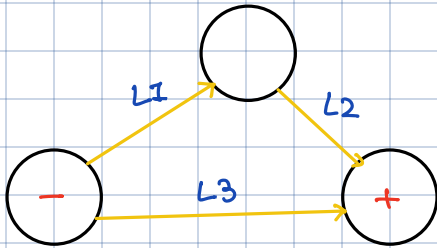
↳ We can take two paths for 'baab'. FA had a unique path.
↳ No path for 'aa' & 'baa' takes to same state but not final. So, both are rejected.

↳ Such machines will be referred as transition graphs (TGs)

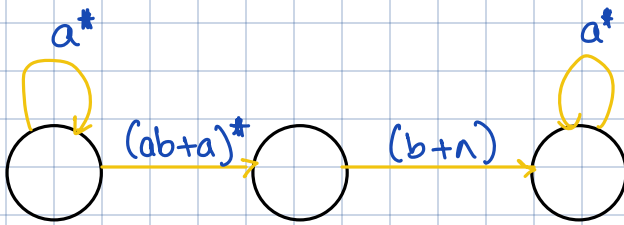
↳ We can have multiple start states now.

↳ Every FA is a TG but not viceversa.

Generalized transition Graph (GTG)

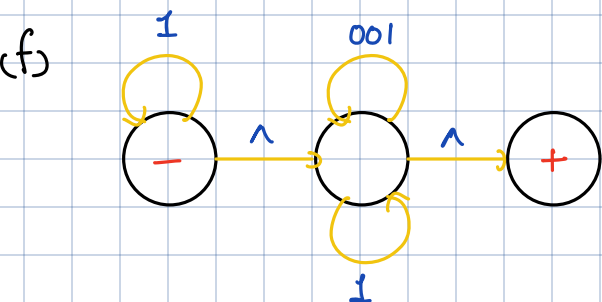
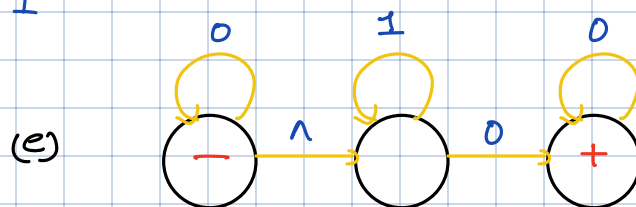
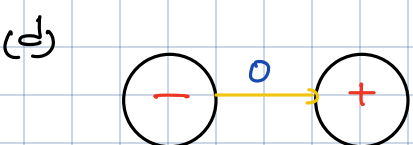
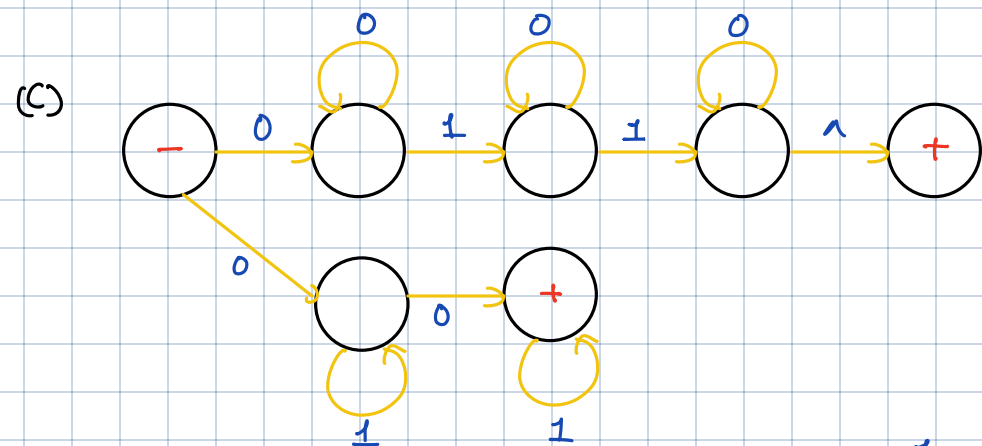
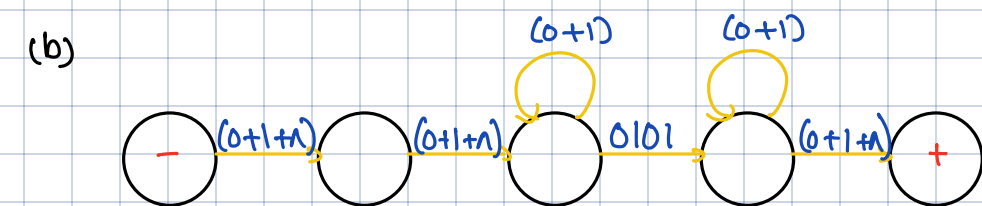
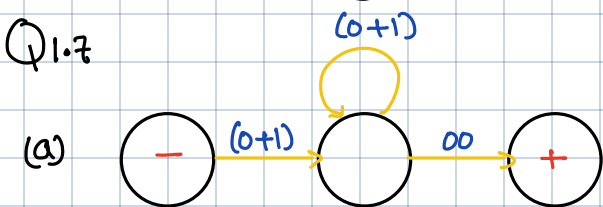


Where $L1, L2, L3$ are Regexes.

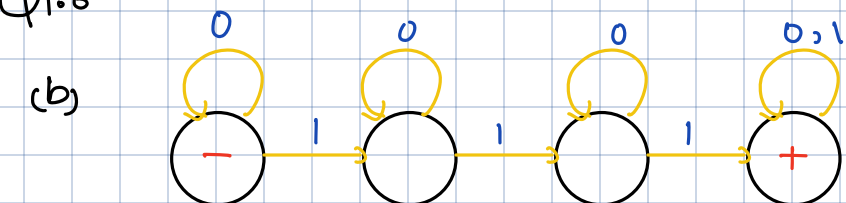


Problems: (Theory of computation)

Q1.7



Q1.6



(d)

